

Predict the Branch

Student Worksheet | CISP 360 - Lecture 06.02: Conditionals

Name: _____

Date: _____

Directions

For each program, predict the exact output before running it. Record the output, then explain which condition or branch caused that result. Compare your reasoning with a partner before checking the program.

Prediction checklist

1.	Evaluate the condition.
2.	Identify the branch that runs.
3.	Ignore statements in branches that do not run.
4.	Write the exact output, including capitalization and spacing.

Example 1

```
int score = 72;

if (score >= 70) {
    cout << "Pass";
}
```

Predicted output:

Why?

Example 2

```
int age = 17;

if (age >= 18) {
    cout << "Adult";
}

cout << "Welcome";
```

Predicted output:

Why?

Example 3

```
int temperature = 32;

if (temperature < 32) {
    cout << "Freezing";
} else {
    cout << "Not freezing";
}
```

Predicted output:

Why?

Predict the Branch

Continue the predictions before running the programs.

Example 4

```
int score = 85;

if (score >= 90) {
    cout << "A";
} else if (score >= 80) {
    cout << "B";
} else if (score >= 70) {
    cout << "C";
} else {
    cout << "D";
}
```

Predicted output:	
Why?	

Example 5

```
int number = 12;

if (number > 10) {
    cout << "A";

    if (number < 20) {
        cout << "B";
    }
}

cout << "C";
```

Predicted output:	
Why?	

Discussion and Debrief

1.	Which condition was evaluated first?	
2.	Which statements were skipped?	
3.	Why did Example 4 stop at B instead of continuing to C?	
4.	In Example 5, how does the nested if depend on the outer condition?	
5.	What would change if the values were exactly on the boundary?	